City University of Hong Kong

CS5487 Machine Learning

Semester A 2020/21
Online Midterm
Time: 2 hours

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1. The midterm has 5 pages including this page, consisting of 3 questions.
2. The following resources are allowed on the midterm:
   - You are allowed a cheat sheet that is **one** A4 page (**single-sided only**) hand-written with pen or pencil.
3. All other resources are not allowed, e.g., internet searches, classmates, other textbooks.
4. Answer the questions on physical paper using pen or pencil.
   - Answer **ALL** questions.
   - Remember to write your **name, EID, and student number** at the top of each answer paper.
5. You should stay on Zoom during the entire exam time.
   - If you have any questions, use the private chat function in Zoom to message Antoni.
6. Midterm submission
   - Take pictures of your answer paper and submit it to the "Midterm Quiz" Canvas assignment. You may submit it as jpg/png/pdf.
   - *It is the student's responsibility to make sure that the captured images are legible. Illegible images will be graded as is, similar to illegible handwriting.*
   - If you have problems submitting to Canvas, then email your answer paper to Antoni (abchan@cityu.edu.hk).

Statement of Academic Honesty

Below is a **Statement of Academic Honesty**. Please read it.

I pledge that the answers in this exam are my own and that I will not seek or obtain an unfair advantage in producing these answers. Specifically,

- I will not plagiarize (copy without citation) from any source;
- I will not communicate or attempt to communicate with any other person during the exam; neither will I give or attempt to give assistance to another student taking the exam; and
- I will use only approved devices (e.g., calculators) and/or approved device models.
- I understand that any act of academic dishonesty can lead to disciplinary action.

I pledge to follow the Rules on Academic Honesty and understand that violations may led to severe penalties.

Name:
EID:
Student ID:
Signature:

(a) Copy the above statement of academic honesty to your answer sheet. Fill in your name, EID, and student ID, and sign your signature to show that you agree with the statement and will follow its terms.

Problem 1 MLE for Geometric distribution [30 marks]

In this problem you will consider the maximum likelihood estimate (MLE) of the parameters of a *Geometric Distribution*.

Suppose we have a Bernoulli random variable with probability π of "success", and probability $1 - \pi$ of "failure". The geometric distribution is the probability that we need x independent Bernoulli trials to see the first occurrence of "success". For example, if the Bernoulli r.v. is a coin and π is the probability of "Heads", then the geometric distribution models the number of times we need to flip the coin to see the first occurrence of "Heads".

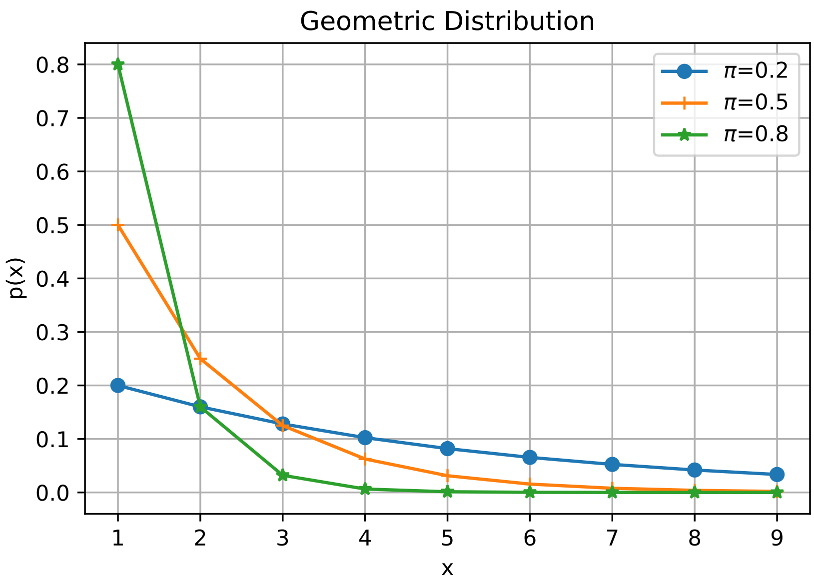
Formally, the geometric distribution is:

$$p(x|\pi) = (1 - \pi)^{x-1}\pi \tag{1}$$

where $x \in \{1, 2, 3, \cdots\}$ is the count (the number of Bernoulli trials), and $0 < \pi \leq 1$ is the parameter (probability of "success"). The mean and variance of the geometric distribution are

$$E[x] = \frac{1}{\pi}, \qquad \mathrm{var}(x) = \frac{1 - \pi}{\pi^2}. \tag{2}$$

Here is a plot of the geometric distribution for different values of π:



Suppose we have a set of N samples of the number of trials, $\mathcal{D} = \{x_1, \cdots, x_N\}$, where $x_i \in \{1, 2, 3, \cdots\}$ is the number of trials for the i-th sample.

(a) [5 marks] Write down the log-likelihood of the data $\mathcal{D}$, i.e., $\log p(\mathcal{D}|\pi)$.

(b) [5 marks] Write down optimization problem for maximum-likelihood estimation of the parameter π.

(c) [15 marks] Derive the MLE for the parameter π.

(d) [5 marks] What is the intuitive interpretation of the derived MLE for π.

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Problem 2 MAP for Geometric distribution [35 marks]

In this problem you will compute the MAP estimate for geometric distribution in Problem 1. Let the prior distribution of π be a Beta distribution,

$$p(\pi) = \text{Beta}(\pi|\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\pi^{\alpha-1}(1-\pi)^{\beta-1}, \tag{3}$$

where $\alpha > 1$ and $\beta > 1$ are parameters of the Beta prior (i.e., *hyperparameters*). $\Gamma(z)$ is the gamma function, which is an extension of factorial to positive real numbers, defined by

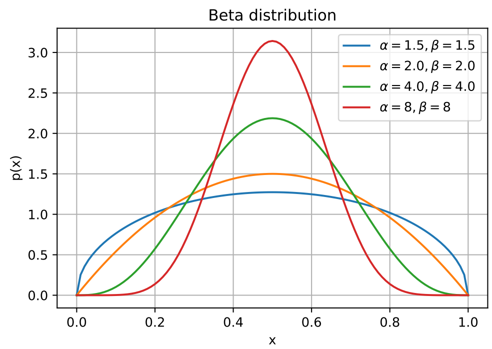
$$\Gamma(z) = \int_0^\infty u^{z-1}e^{-u}du. \tag{4}$$

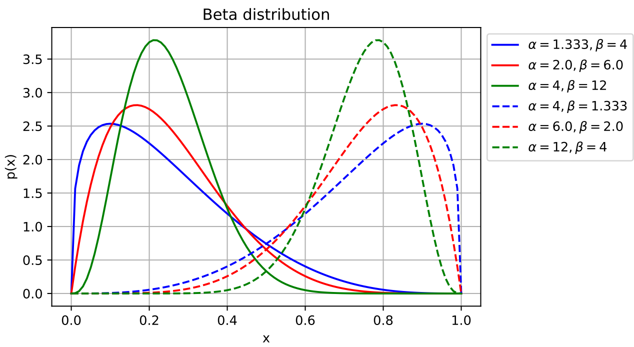
The Gamma function obeys the relation $\Gamma(z+1) = z\Gamma(z)$. Furthermore, $\Gamma(1) = 1$, and $\Gamma(z+1) = z!$ when z is a non-negative integer.

The mean, variance, and mode of $\text{Beta}(\alpha, \beta)$ are:

$$E[\pi] = \frac{\alpha}{\alpha+\beta}, \qquad \text{var}(\pi) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}, \qquad \text{mode}(\pi) = \frac{\alpha-1}{\alpha+\beta-2}. \tag{5}$$

The following plot shows some examples of the Beta distribution for different parameters (α, β):





(a) [5 marks] Write down the optimization problem for MAP estimation of π using a Beta prior with known hyperparameters (α, β).

(b) [15 marks] Derive the MAP estimator for the parameter π using the Beta prior.

(c) [10 marks] Compare this MAP estimator with the ML estimator derived in Problem 1. What is the intuitive interpretation of the MAP solution with regards to the ML solution?

(d) [5 marks] What is the intuitive effect on the MAP estimate as the number of samples N increases?

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Problem 3 Bayesian estimation for Geometric distribution [35 marks]

In this problem you will derive the Bayesian estimate for the Geometric distribution with Beta prior, using the same setup as Problems 1 and 2.

(a) [5 marks] Write down the posterior distribution of the parameters, $p(\pi|\mathcal{D})$, in terms of the prior and likelihood function.

(b) [15 marks] Derive the form of the posterior distribution $p(\pi|\mathcal{D})$ in terms of the hyper-parameters (α, β) and data $\mathcal{D}$.

(c) [5 marks] What is the intuitive interpretation of the derived Bayesian estimate in Problem 3(b)?

(d) [10 marks] What happens to the posterior as the number of samples N increases?

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